

The Technical Specifications Manual

This is the comprehensive, official **Internship Program Documentation & System Specification Plan** issued on behalf of **Arabian Arrow Supply and Services Co Ltd (ARACO), Dammam, Saudi Arabia**.

This document consolidates all your operational workflows, the 9-stage verification pipeline, multi-company data structure requirements, and the core AI milestones into one unified blueprint. You can print or email this directly to Fariz Sameer as his official framework.

Internship Project Blueprint & Technical Specifications

Company : Arabian Arrow Supply and Services Co Ltd (ARACO)

Authorized By : Arshad Mohamed, CEO

Intern Name : Fariz Sameer, B.Tech CSE, NIT AP, farizsameer@gmail.com

Department : Systems Engineering

Internship Duration : 3 Weeks (Concluding July 27, 2026)

Project Title : AI-Powered Multi-Company Daily Task Management System (TMS)

1. Project Objective & Core Mandate

Welcome to Arabian Arrow Supply and Services Co Ltd (ARACO). Over the next three weeks, you will design, develop, and deploy a functional, lightweight, enterprise-grade **Daily Task Management System (TMS)**.

Because ARACO oversees operations intersecting multiple corporate entities, this software must seamlessly track cross-company workflows. Furthermore, implementing cutting-edge generative AI capability is a core priority; the system must utilize AI to aggregate raw technical outputs into clear, conversational management summaries.

The Technical Framework

To guarantee delivery before the final presentation deadline, you will utilize a rapid micro-framework stack:

- **Interface & Frontend:** Python (via the **Streamlit** library) for clean, responsive UI controls.
- **Database Engine:** **SQLite** for relational, transactional storage.

- **Artificial Intelligence: Google GenAI SDK** (gemini-2.5-flash) for automated executive summaries.

2. Structural Requirements (Database Schema)

Every task transaction entry created or modified within the database must store and map the following strict data attributes. You are required to enforce database constraints preventing critical fields from being submitted blank:

- **Unique Serial Number (ID):** An auto-incrementing unique integer assigned to each task.
- **Task Date** (task_date): The date the record was initiated.
- **Company Name** (company_name): A dropdown classification selector to isolate which sister/inter-company the task belongs to.
- **Task Details** (task_details): A text description text area specifying the technical or operational work required.
- **Assigned Member** (assigned_to): The team member executing the work.
- **Timeline/Deadline** (deadline): A date-picker selection specifying the target completion date.
- **Assignor Remarks** (assignor_remarks): A running log of feedback, review notes, or directives appended by the manager.
- **Assignee Notes** (assignee_notes): A running log of progress details or blockers appended by the employee.
- **Task Status** (status): Tied directly to the workflow validation rules below.

3. The 9-Stage Status Workflow & Color Rules

The UI dashboard must visually flag task priorities and lifecycle states utilizing explicit color coding. You must implement the following state logic:

Stage Status	UI Badge Color	Permission & Verification Logic
 Urgent	Bright Red	Assigned by management for immediate priority items.
 In Progress	Amber / Yellow	Active state selected by the Assignee while working.

Stage Status	UI Badge Color	Permission & Verification Logic
 Pending	Blue	Waiting on external variables, clients, or inputs.
 For Discussion	Orange	Internal flag requiring a meeting or leadership clarification.
 Not Ready	Light Grey	Future pipeline item not yet initiated.
 Finished	Purple	Set by Assignee. Signals that work is complete and ready for review.
 Completed	Green	Set ONLY by Assignor. Manager reconfirms work and officially closes the task.
 Closed to Reopen	Dark Charcoal	Set by Assignor if the finished work fails verification.
 Cancelled	Muted Grey	Strikethrough style for discarded or invalidated entries.

Two-Step Verification Guardrail

You must program a strict permission logic barrier into the interface layout:

1. Users operating under an **Assignee** profile are blocked from selecting or transitioning any task to  **Completed**. They can only advance a task up to  **Finished**.
2. A specialized alert view must be rendered on the Manager dashboard highlighting all tasks sitting at  **Finished**, prompting the **Assignor** to inspect, append comments to assignor_remarks, and manually execute the final transition to  **Completed** or  **Closed to Reopen**.

4. Artificial Intelligence Integration Mandate

Implementing AI technology into the manager infrastructure is a critical component of this project. You will build an analytical utility powered by gemini-2.5-flash.

- **The Core AI Feature:** Build an automated "Executive Daily Digest" engine.

- **Functional Logic:** The system must capture all active logs across all companies for a designated date, feed the structured text chunk securely to the Gemini API, and return a beautifully structured, 3-bullet executive summary evaluating:
 1. Overall cross-company operational velocity.
 2. Critical bottlenecks, high-priority issues, or unresolved ● **Urgent** items.
 3. Operational targets for the next business day.
- **Security Rule:** API Keys must be dynamically accessed via system environment variables or hidden framework secrets. **Never hardcode credentials.**

5. Accelerated 3-Week Project Timeline

Your progress and final internship evaluation will track precisely against this execution timeline:

Architecture & Relational Schema

Week 1: July 6 – July 12

Initialize local code repository environment. Build the SQLite relational schema enforcing the data attributes, tracking fields, and multi-company parameters. Construct the base Streamlit view layout with color-coded badge tables.

Verification Logic & AI Pipeline

Week 2: July 13 – July 19

Program the Two-Step Verification interface (Assignee 'Finished' restriction versus Assignor 'Completed' authorization). Install the Google GenAI SDK, design prompt profiles, and engineer the automated cross-company daily summary generator.

Validation, Handover & Final Presentation

Week 3: July 20 – July 27

Exhaustively test edge cases (preventing empty string inputs, ensuring formatting remains cohesive during modifications). Write the technical product documentation (README.md).

Live software demonstration and project presentation to the CEO on July 27.

6. Project Evaluation Criteria

Your work will be scored on July 27th across four performance parameters:

- **Functional Stability (40%):** Does the app reliably execute data entry, enforce status colors, restrict permissions, and query the AI backend without crashing?
- **Code Architecture & Documentation (20%):** Is the script modular, cleanly structured, and well-documented so it can be added upon as the system progresses?

- **Punctuality & Tracking (20%):** Did you systematically advance through the weekly sprint milestones?
- **Professional Presentation (20%):** How articulately can you showcase your software's user flow and immediate utility to company leadership during the final review?

Certification Gateways

Official credentialing at the close of this internship is strictly tied to performance execution metrics:

- **Score $\geq$ 70%:** Upon reaching a successful completion score of 70% or above, the company will award the official **“Internship Certificate of Completion”**.
- **Score < 70%:** If the project remains incomplete or fails to meet functional requirements, but the internship timeline was fully attended and supported, the company will issue an **“Internship Appreciation of Participation”**.

Corporate Authorization

A Note from the CEO: > *"Fariz, at ARACO, we believe in creating software solutions that are sharp, efficient, and immediately useful. This Task Management System represents a critical internal tool designed to give us structural oversight across multiple operations. Your goal over these 3 weeks is to construct a stable database foundation, build an immaculate color-coded workflow, and leverage AI to make monitoring simple. Give this project your absolute best focus, clear the 70% evaluation bar, and secure your Certificate of Completion. We look forward to your live application demonstration on July 27."* >

Arshad Mohamed > Chief Executive Officer

Arabian Arrow Supply and Services Co Ltd (ARACO)

Intern Signature: _____ **Date:** ____ / ____ / 2026

Management Signature: _____ **Date:** ____ / ____ / 2026